

1. Introduction

My solution is built on PromptIR, an all-in-one blind image restoration framework that conditions a shared backbone on a small pool of learnable prompts, letting a single network handle both rain and snow without any explicit type labels. I implement it with Restormer backbone with MDTA channel attention, GDFN gated feed-forward, and three Prompt-Generation Blocks (37.5 M parameters). Training uses AdamW with cosine learning-rate annealing, Charbonnier loss, automatic mixed precision (AMP), and standard augmentations (random 128×128 crops, flips, rotations). At inference I apply 8-way self-ensemble (TTA). Final system reaches 30.05 dB PSNR on the public leaderboard.

Github link: <https://github.com/mses0424/Visual-Recognitionusing-Deep-Learning-hw4>

2. Method

2.1 Data Preprocessing

To monitor training, I hold out the last 50 images of each type (100 in total) as a validation set, leaving 3,100 images for training. Validation PSNR is computed on full-size images, while training uses random patches. The validation split is deterministic so that scores are comparable across training sessions and resume checkpoints.

Training-time augmentation:

- (1) Random 128×128 crop from each full-size image.
- (2) Random horizontal flip (probability 0.5).
- (3) Random vertical flip (probability 0.5).
- (4) Random 90° rotation ($k \in \{0, 1, 2, 3\}$ chosen uniformly).
- (5) Pixel values are normalized to [0, 1] by dividing by 255.
- (6) Images smaller than 128×128 are reflect-padded before cropping.

Since the model uses multiple down-sampling and up-sampling operations, test

images whose height or width is not a multiple of 16 are reflect-padded on the right and bottom before inference.

2.2 Model Architecture

I follow the official PromptIR design without architectural modifications. The model is a U-shaped encoder-decoder of transformer blocks, augmented with three Prompt-Generation Blocks (PGBs) that inject input-adaptive prompts at three points in the decoder. Each transformer block combines a channel-wise self-attention (MDTA) and a gated feed-forward network (GDFN), both inherited from Restormer.

Main design:

- (1) Overlapping patch embedding (3×3 convolution): $\text{RGB} \rightarrow \text{C} = 48$ channels.
- (2) Three encoder stages with $[4, 6, 6]$ transformer blocks and channel multipliers $\times 1, \times 2, \times 4$. Pixel-Unshuffle is used for down-sampling.
- (3) Latent stage of 8 transformer blocks at $\times 8$ channels.
- (4) PGB-3 (Third Prompt-Generation Block) injects a prompt at the latent stage; a fusion transformer block + 1×1 convolution reduces the channel count back.
- (5) Three decoder stages (mirroring the encoder) up-sample with Pixel-Shuffle and concatenate skip features. PGB-2 and PGB-1 inject prompts before stage 2 and stage 1 respectively.
- (6) A final 3×3 convolution produces the residual that is added to the input image.

2.3 Hyperparameter Settings

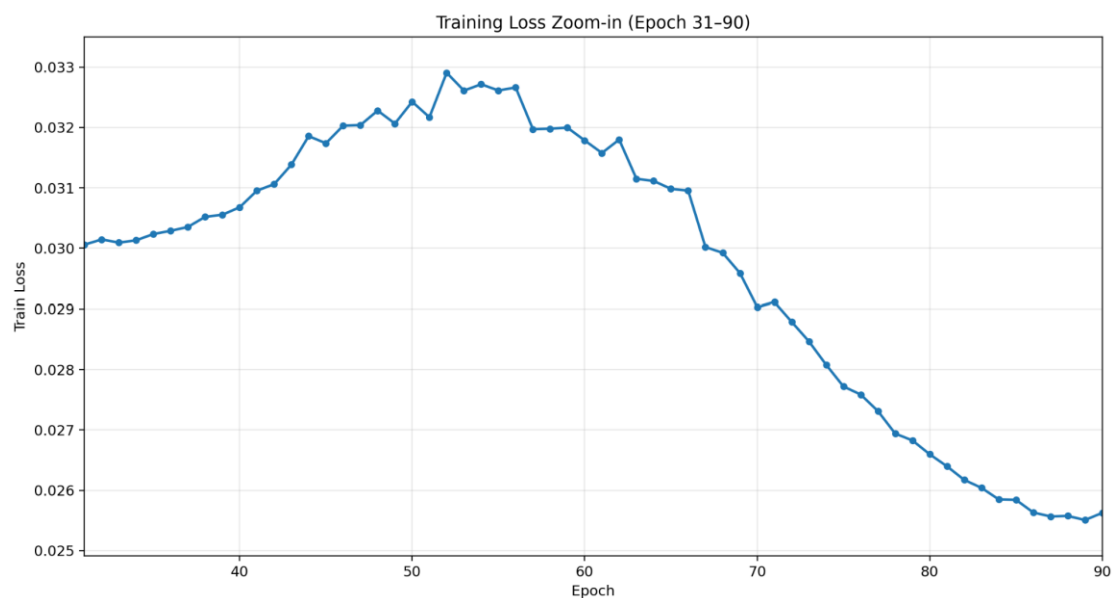
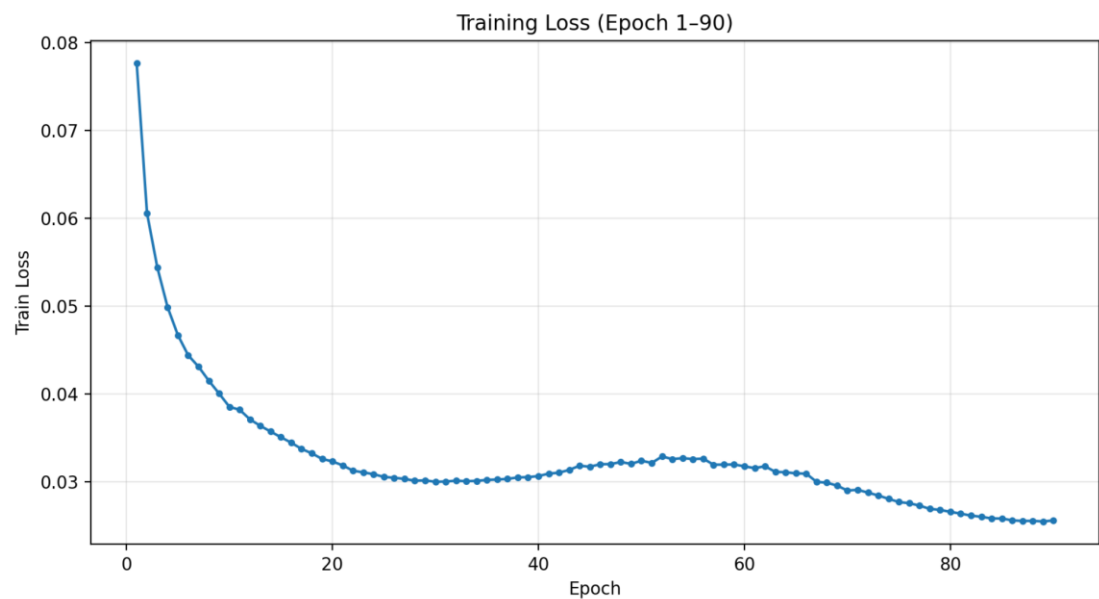
Patch size	128*128
Batch size	4
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\text{wd} = 1\text{e-}4$
Initial learning rate	$2\text{e-}4$
Minimum learning rate	$1\text{e-}6$
LR schedule	Cosine annealing
Total epochs	90
Gradient clipping	1.0 (L2-norm)
Loss function	Charbonnier, $\epsilon = 1\text{e-}3$

3. Results

Training loss curve

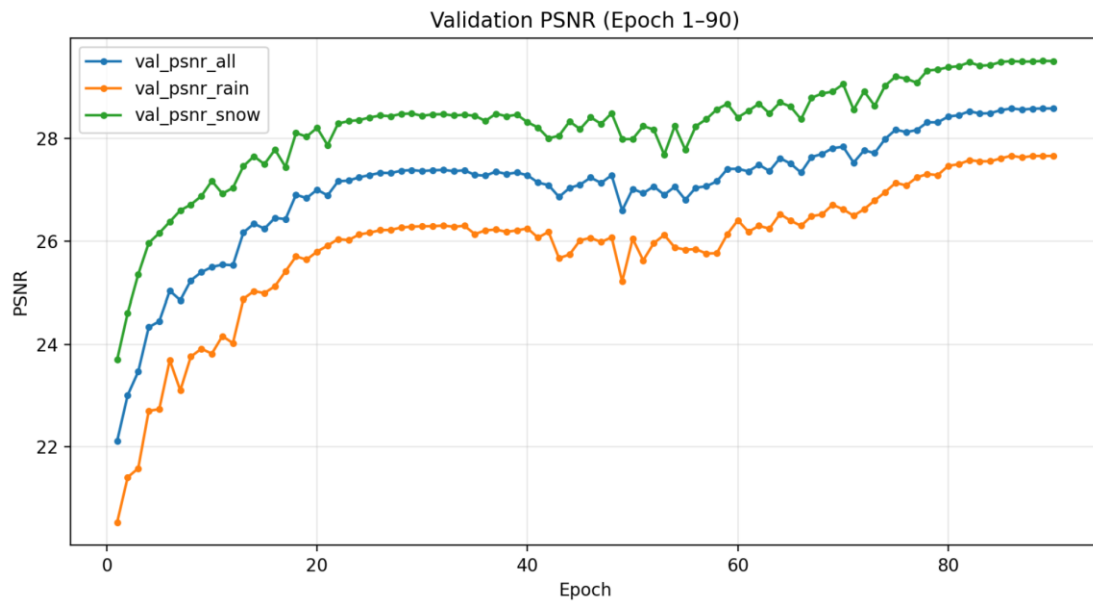
The training loss decreases from 0.077659 to 0.025626, showing that the model learns effectively during training. The loss drops quickly in the early epoch.

Although there is a small fluctuation in the middle stage, it decreases again after around epoch 60 and becomes stable near the end. This indicates that the model gradually converges without obvious training instability.



Validation PSNR curve

Metric	Epoch 1	Epoch 90	Best Value
Validation PSNR All	22.1144 dB	28.5852 dB	28.5862 dB at epoch 86
Validation PSNR Rain	20.5286 dB	27.6601 dB	27.6640 dB at epoch 86
Validation PSNR Snow	23.7002 dB	29.5102 dB	29.5115 dB at epoch 89



The model performs better on snow removal than rain removal, since the final snow PSNR reaches 29.5102 dB, which is higher than the rain PSNR of 27.6601 dB. This suggests that snow degradation may be easier for the model to restore.

However, rain PSNR shows a larger improvement, increasing by about 7.13 dB, while snow improves by about 5.81 dB. This suggests that rain restoration is more difficult than snow restoration, even though the model still improves significantly on rain images.

The best overall PSNR appears at epoch 86, and the result at epoch 90 is almost the same. This shows that the model becomes stable near the end and does not show obvious overfitting.

4. References

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- [3] I. Loshchilov and F. Hutter. "Decoupled Weight Decay Regularization."
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